RIDDHI KUMARI SINGH

GenAI | MLOps & LLMOps Engineer | Platform Architect Building Scalable Generative AI, Computer Vision & ML Solutions AWS • Azure • On-premise | Leading 2–3 Engineer Teams

PROFILE

GenAI & MLOps Engineer with 4+ years in enterprise projects. Expert in Generative AI / GenAI applications using LLMs (Llama, GPT, Stable Diffusion), LLM fine-tuning, RAG, computer vision, NLP, and scalable MLOps/LLMOps pipelines with PyTorch, TensorFlow, Hugging Face, Kubeflow, SageMaker, Docker, Kubernetes, AWS & Azure. Led 2-3 engineer teams to deploy production AI/ML solutions, optimize workflows, and deliver high-impact results for large enterprises.

CONTACT

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EMAIL

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LOCATION

Bangalore, Karnataka

EDUCATION

Master of Technology - Cloud Computing
Indian Institute of Technology, Patna
Mar 2023 – Aug 2025

Bachelor of Technology - Information Technology
Banasthali Vidyapith, Jaipur
July 2017 – June 2021

LINKEDIN PROFILE
https://www.linkedin.com/in/riddhi-kumari-singh/

WEBSITE
https://riddhiksingh8.github.io/riddhiksingh/

GITHUB LINK
https://github.com/riddhiksingh8

CERTIFICATES
M001: MongoDB Developer
M100: MongoDB SQL Pro
Python with Data Science Machine Learning

WORK EXPERIENCE 4.6+ years

Ernst & Young LLP
Senior Consultant
Apr 2024– Present

- Led a team of 2-3 engineers in Platform & Solution Engineering, focusing on MLOps and Data Architecture for Gen AI initiatives.
- Developed scalable data pipelines and microservices using Python, TypeScript, FastAPI, and Flask, emphasizing Gen AI model deployments.
- Managed CI/CD pipelines on Azure and defined architectural best practices.
- Awarded "Excellence Award" for exceptional project delivery.

SecLogic INC– Software Development Engineer
Jan 2023– Mar 2024

- Guided a small team of 2 engineers in Software Development, Cybersecurity, and Data Analysis with a focus on Gen AI-enhanced security solutions
- Designed and implemented secure APIs for integrating threat intelligence into platforms, incorporating Gen AI for predictive analytics.
- Led the development of security automation scripts, streamlining vulnerability management and patch deployment using MLOps principles.

Theecode Technologies Pvt Ltd – Software Development Engineer
Oct 2021– Dec 2022

- Supervised a team of 3 in Development, Data Analysis, and Computer Vision projects, integrating Gen AI for advanced insights.
- Applied computer vision algorithms for accurate motion tracking and analysis, enhanced with generative AI models.
- Built and managed data pipelines for processing sensor data, demonstrating expertise in system scalability, data-driven insights, and MLOps for real-time AI deployments.

SKILLS

ENVIRONMENTS
AWS Azure

MLOPS PLATFORMS
Databricks Sagemaker Kubeflow Sagemaker Studio Microsoft Fabric Domino Google Vertex Azure AI Foundry

CONTAINERS
Docker Singularity Amazon Elastic Container Service ECS Elastic Kubernetes Service EKS Google Kubernetes Engine Azure Kubernetes Service

LANGUAGES & Framewiorks
Python Javascript React JS Node JS Express JS Play Framework Flask FAST API Django bash script shell AWK Sed Gunicorn Uvicorn REST APIs Nginx React.js Redux Material UI Charting Libraries, *System Architecture*

INFRASTRUCTURE AS CODE
Terraform Cloud Formation

VISUALIZATION
Matplotlib seaborn Tableau Grafana Quicksight D3 JS

OS
Linux Unix Windows mac

DATABASES & STORAGE
PostgreSQL MongoDB Neo4j Redis Azure SQL Database Data Lake AWS S3 Google cloud storage Azure blob RabbitMQ Celery

NETWORKING
VPC Subnet – Public/Private DNS IPSec NAT Gateway Internet Gateway Transit Gateway AutoScaling IPv4 IPv6 Ngnix

AI/ML Pytorch Sklearn TensorFlow Hugging Face NLP Computer Vision, GenAI

WORKFLOW & MONITORING & CI/CD
GitHub Actions DataDog Airflow Step Functions Kubeflow Pipeline AWS CloudWatch

SOME LEADING BRANDS WORKED WITH

WORKED IN ENTERPRISE PROJECTS SUCH AS PLATFORMS FOR LARGE ENTERPRISE, AIOPS & MLOPS, ML Solutions, Computer Vision Project and Gen AI Projects.

MLOPS Platform	Gen AI Project	MLOPS Domain Implementation
• **MLOPS Platform** - Implemented a comprehensive CI/CD pipeline as a Service Catalog product utilising AWS services including RDS, EMR, Sagemaker Studio, ECR, and EKS. Leveraged federated IAM roles for secure cross-account access between a master account and multiple slave accounts, enhancing scalability for a large user base. Automation was fully achieved using CloudFormation and Terraform, optimising deployment efficiency and resource management across the organisation. • **Outcome & Impact:** This streamlined architecture significantly reduced deployment times and operational overhead, while improving security and compliance across accounts. The solution enabled seamless scalability and agile management of resources, leading to enhanced productivity and cost-efficiency in cloud operations.	• **Gen Ai Project** - Developed an intelligent document processing solution integrating API endpoints with a vector database for enhanced query performance and semantic search. Deployed Azure OpenAI's GPT-3.5 and GPT-4 models to analyze and retrieve insights from scanned PDF documents, saving model checkpoints securely in S3 for version control and disaster recovery. • Created API interfaces linked to a vector database for efficient indexing and querying of document content, enabling quick, context-aware responses to user questions. • **Outcome & Impact:** This setup improved response accuracy for document-based queries, allowing users to interactively and quickly access information. The integration of GPT models with vector search capability increased productivity and enhanced the document retrieval experience for end-users.	• **MLOPS project** - Directed a team of 2 in implementing an advanced system for hyperparameter tuning using AWS Sagemaker, orchestrated through AWS Step Functions. This system automatically triggers tuning jobs when specific files are uploaded to a designated S3 bucket, facilitated by AWS CloudWatch notifications. The entire process, including the setup and execution of the hyperparameter tuning jobs, is automated using an AWS CloudFormation template. • **Outcome & Impact:** This configuration ensures optimal model selection by leveraging continuous and automatic hyperparameter adjustments, which enhance model performance and accuracy. The automation of this process minimizes manual oversight and accelerates the model training and deployment cycle, leading to more efficient resource utilization and improved outcomes in predictive performance.

Cloud Infrastructure / DevOps	Data Engineering / NLP	Full-stack Engineering / Risk Intelligence
Summary - Developed a containerized microservices-based platform for real-time news and supplier screening using FastAPI, Node.js, React, and Docker. The system is deployed on an Azure VPS with a centralized Nginx reverse proxy and managed via docker-compose, enabling modular deployment and secure inter-service communication through .env files. • **Outcome & Impact**: The project delivered a modular architecture, enabling each microservice to be independently developed, scaled, and debugged in line with best practices for distributed systems. Through containerization with Docker and fast and zero-downtime deployment on an Azure VPS, complete with internal service discovery. A seamless real-time data flow was established by integrating a web scraper with backend orchestration, ensuring timely updates for downstream analytics. All services are unified under a single public IP. The use of .env-based configuration, strict dependency pinning, and controlled proxy rules ensured a secure and maintainable deployment. Overall, the platform enabled automated supplier and news screening, significantly improving operational efficiency, reducing manual workload, and enhancing compliance with risk protocols.	• **Summary** – Designed and implemented a Python-based real-time scraping engine to extract brand sentiment data from diverse digital platforms including social media, news websites, forums, and blogs. Led a team of 2 in enabling continuous tracking of public sentiment, identifying emerging trends and brand discussions as they happen using Gen AI NLP models. By optimizing the scraping algorithms, refining HTTP request handling, and streamlining resource usage, the platform delivers high-frequency and accurate data ingestion. An intuitive dashboard was also built to visualize sentiment insights, topic frequency, and brand comparisons. The solution further integrates an NLP pipeline to analyze and classify each mention into positive, negative, or neutral categories. • **Outcome & Impact:** This initiative led to a 5–6% boost in scraping efficiency, ensuring timely capture of relevant data across platforms. The NLP integration enhanced brand sentiment understanding, providing actionable insights for marketing and PR teams. The custom dashboard empowered stakeholders to rapidly identify shifts in public perception and respond with agility, significantly reducing dependency on manual brand monitoring.	• **Summary** - Engineered a full-stack web application for GSK's 3rd-party risk monitoring platform, utilizing Flask for the backend and Next.js for the frontend. Led a team of 3 in including a real-time data scraping pipeline integrated with external sources like RocketReach. The solution featured dynamic risk scorecards and a deep-dive analysis page to monitor vendor compliance and financial risks, enhanced with Gen AI for predictive scoring. Ensured seamless cloud deployment for high availability and built-in scalability, alongside advanced error-handling and data validation mechanisms. • **Outcome & Impact:** The system transformed GSK's manual risk assessment into a real-time, data-driven process. Risk managers gained the ability to proactively monitor third-party entities, with instant access to risk scores, compliance flags, and financial health data. This automation led to a notable reduction in decision-making time, improving overall operational agility and regulatory alignment.

• Generative AI / Cloud nfrastructure	• Data Engineering / Computer Vision	• Cloud Security / SaaS
• **Summary** - Architected and deployed a conversational AI chatbot that transforms textual prompts into high-resolution images using a Stable Diffusion model. Led a team of 2 in building an interactive frontend, integrating a natural language interface, and containerizing the backend using Docker for seamless deployment. The chatbot was capable of understanding complex user descriptions and transforming them into visually coherent outputs in seconds. **Outcome & Impact**: The chatbot enabled natural language-driven image generation, creating a bridge between user creativity and visual AI. Its modular design and Docker-based deployment made it portable and easy to scale. The solution demonstrated how NLP and generative AI could be harmoniously combined to deliver educational, creative, and commercial value.	• **Summary** - Led backend development for EXAR, an augmented reality fitness application, guiding a team of 3. Parsed over 10,000 XML files using Python (BeautifulSoup, Pandas) to derive structured datasets. Implemented Dynamic Time Warping (DTW) to analyze real-time body motion and exercise patterns, integrated with Gen AI for enhanced pattern recognition. The platform was delivered through a mobile app using React Native, microservices, and scheduled jobs in AWS EC2 via Cron. Analyzed and visualized performance trends via Excel dashboards. • **Outcome & Impact**: EXAR achieved a 90% increase in user exercise efficiency by leveraging DTW-based motion analysis. Users received instant feedback on their form and consistency, revolutionizing their workout experience. The app became a pioneering solution in fitness tech, proving the viability of real-time posture analysis with intelligent feedback.	• **Summary** - Played a key role in building SecLogic's cloud-native cybersecurity PaaS, featuring two core offerings: CYBERG SHIELD and ORO. Led a team of 2-3 in combining Python (Django), React, Postgres, and AWS to assess digital risk. Advanced components included vulnerability analysis, misconfiguration detection, and third-party risk quantification using thousands of dynamic data points, augmented with Gen AI models. Visual insights were delivered through D3.js dashboards, with rigorous OAuth-based security and CI/CD pipelines. • **Outcome & Impact**: The platform delivered a 60% reduction in cybersecurity incidents for enterprise clients in its first year. By providing real-time, contextual risk scoring, SecLogic empowered security teams to make faster and more informed mitigation decisions. It became a benchmark solution for organizations seeking end-to-end digital risk visibility.

PROFESSIONAL ACHEIVEMENTS

- Got **Emerging Extraordinaire** award in first 6 months for my contributions in projects within a Fortune 500 company in EY
- Got **Benepik Plus Gift Card** / amazon coupon for outstanding performance at GSK.
- Participated in www.Guvi.in code kata competition and got first place for Solving Puthon puzzle's and for later use stored in GitHub - https://github.com/riddhiksingh8 They offered me to recommendation of suitable internship's and jobs Interview's.
- Became first Microsoft Student Ambassador where Hosted 6 Events on Azure